

Doug Evan's Preemptive Strike Rate Based on Race

Aryan Arora and Muno Siyakurima

2/23/2022

Introduction:

This study examines the preemptive strike rate of Mississippi District Attorney, Doug Evans. A preemptive strike gives an attorney the right to, during jury selection, reject a potential juror without cause. The data used in this study comes from 89 trials in which Doug Evans was the primary prosecutor. In our study we use the data collected to examine the relationship between our predictor variables and the odds of a juror being preemptively struck by Evans. Our predictor variables are the following:

- Whether the juror has been accused of being involved in criminal activity
- Whether the juror is black, with a reference level of the juror being white
- Whether the juror has a family member or close friend who has been accused of being involved in criminal activity
- Whether the juror expressed reservations about imposing the death penalty
- Whether the juror has prior familiarity with the defendant
- Whether the juror was the same race as the defendant
- Whether the juror has a family member or close friend who is in law enforcement'

Our focus is to use logistic regression to determine whether Evan's use of the preemptive strike denied black jurors the opportunity to appear in court and prevented defendants from exercising their constitutional right to be tried by a jury of their peers. The data can be found at(<https://tinyurl.com/mrx7wht4>).

Results:

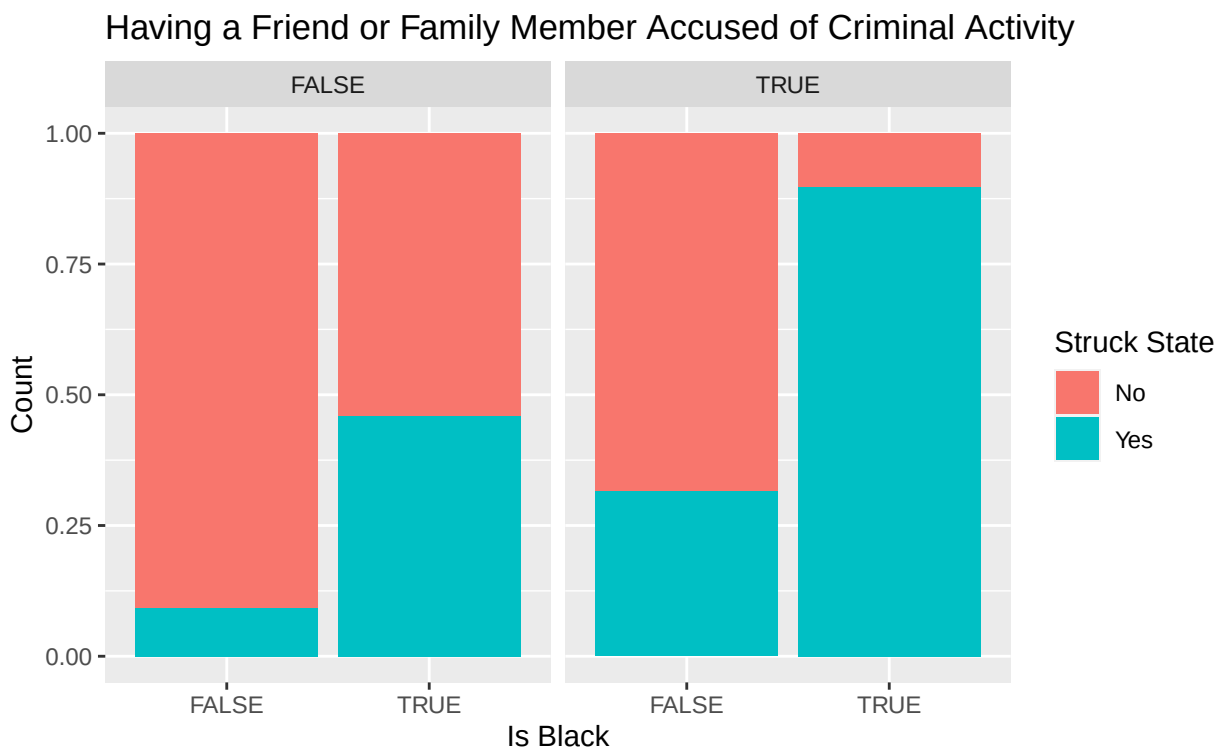
We do not believe that the model presented by the APM is adequate. In examining the graph of the proportion of people struck dependent on whether they were black and if they have a friend or family member accused of engaging in criminal activity, we found there to be evidence of an interaction between being black and having a friend or family member who has been accused of criminal activity. This interaction was verified through individual t-tests of the coefficients which provided the p values 0.028081 and 0.021352 respectively—providing strong evidence of a non zero interaction coefficient for both at a 5% significance level. Below is a table showing the estimated coefficients and p values for the predictor variables in our model. The p values provide evidence of non-zero coefficients, the estimates provide evidence of each predictors contribution to the odds of being struck, and the standard errors provide information about the standard error of each coefficient. Both the coefficient estimates and standard error estimates have been unlogged.

Coefficients:

Estimate	Std. Error	z value	Pr(> z)	Exponentiated Estimate	Exponentiated Std. Error	p value
(Intercept)				0.09333128	1.10871313	<2e-16
Same Race				1.39947897	1.15096417	0.016826
Has Family in Law Enforcement				0.57868326	1.1775076	0.000818
Has Reservations about Death Penalty				6.32303934	1.79247976	0.001576
Knows Defendant				3.78596194	1.24844653	1.97e-09
Has Been Accused of Crime				12.376506	1.71755196	3.31e-06
Is Black				6.10372964	1.15777493	<2e-16
Has an Accused Family Member/Friend				4.87343974	1.23059772	2.28e-14
Black + Has an Accused Family/Friend				2.16820579	1.44947294	0.037092

We chose to include the race variable instead of the defendant race variable because including the defendant race variable did not improve our model. The individual coefficient t-test p-values for the 3 defendant race variables were 0.176801, 0.842523 and 0.367357 respectively. These insignificant p-values indicated that these variables did not improve our model and thus, we maintained the race variable in our model.

The graph below provides evidence of an interaction between being black and having a friend or family member who has been accused of criminal activity because change in proportion of people struck between people in the first and third bar plot (someone who is white and does not have a friend or family member accused of criminal activity and someone who is white and does have a friend or family member accused of criminal activity) differs from the difference in proportions between the second and fourth bar plot (someone who is black and does not have a friend or family member accused of criminal activity and someone who is black and does have a friend or family member accused of criminal activity). This difference in slopes between the proportions indicates that the effect of having a friend or family member who has been accused of criminal activity affects blacks and whites differently when examining their odds of being struck.



Using our regression model, we found that the odds of being struck were, on average, 6.1 times greater for someone who is black and compared to someone who is not, fixing all other variables. We are 95% confident that, after fixing for all other variables, the odds of being struck are between 4.6 to 8.1 times greater for someone who is black compared to someone who is white. We are 95% confident that a black person who had a friend or family member who was accused of criminal activity, was between 6.5 and 26.9 times more likely to be struck as compared to a white person with a family member who was accused of criminal activity. The mean odds of a black person who had a family member who was accused of criminal activity being struck was 13.2 times greater than for a white person.

Discussion:

Our results indicated that Evan's use of the preemptive strike denied black jurors the opportunity to appear in court and prevented defendants from exercising their constitutional right to be tried by a jury of their peers. We are 95% confident that Evans dismissed black jurors at a rate 4.6 to 8.1 times greater than for white jurors. We are also 95% confident that Evans preemptively dismissed black jurors who had a friend or family member who had been accused of engaging in criminal activities at a rate 6.5 to 26.9 times greater than for white jurors who had a friend or family member who had been accused of engaging in criminal

activities.

One main limitation in this study was when looking for interactions between whether the juror was black and the other predictor variables, some of the plots contained only one observation and therefore, demonstrated an interaction between the two variables that we cannot appropriately verify.

We also would have loved to explore some other variables in the dataset and how they affected the odds of a state strike. There were additional interactions we saw when looking at the graphs other predictor variables, but we ultimately chose not to include these variables either because the p values of their individual t tests did not provide a statically significant result at a 5% significance level or because including them created a non-significant p value for the predictor variable (the non-interaction term). With further time and a project of a larger scope, we would consider relaxing our significance level and reconsidering some of these interactions. We would also want to consider whether we see similar rates of striking among other prosecutors or similar rates of striking from Doug Evans towards other cultural groups.

R Appendix:

```

----
title: "Doug Evan's Preemptive Strike Rate Based on Race"
author: "Aryan Arora and Muno Siyakurima"
date: "2/23/2022"
output: pdf_document
----

```{r setup, include=FALSE}
apm <- read.csv("http://people.carleton.edu/~apoppick/ClassData/apm.csv", header = TRUE)
library(ggplot2)
library(tidyverse)
library(broom)
```

```{r, include=FALSE}
apm.lm <- glm(data = apm, struck_state_bin ~ accused + is_black + fam_accused +
death_hesitation + know_def + same_race + fam_law_enforcement + know_wit +
fam_crime_victim + crime_victim + prior_info + gender, family = binomial)

```

```{r, include=FALSE}
apm.lm2 <- glm(data = apm, struck_state_bin ~ accused + is_black + fam_accused +
death_hesitation + know_def + same_race + fam_law_enforcement, family = binomial)

summary(apm.lm2)
```

```{r, include=FALSE}
apm.lm3 <- glm(data = apm, struck_state_bin ~ same_race + fam_law_enforcement +
death_hesitation + is_black * accused + is_black * fam_accused + is_black * know_def,
family = binomial)

summary(apm.lm3)

apm.lm4 <- glm(data = apm, struck_state_bin ~ same_race + fam_law_enforcement +
death_hesitation + know_def + accused + is_black * fam_accused, family = binomial)

summary(apm.lm4)
```

```{r, include=FALSE}
library(patchwork)
Accused <- ggplot(apm, aes(x = is_black, fill = struck_state)) +
geom_bar(position = "fill") +
facet_wrap(~ accused) +
 labs(x = "Is Black", y = "Count", fill = "Struck State", title = "Accused") +
 scale_fill_discrete(labels = c("No", "Yes"))

Fam_Accused <- ggplot(apm, aes(x = is_black, fill = struck_state)) +
geom_bar(position = "fill") +
facet_wrap(~ fam_accused) +
 labs(x = "Is Black", y = "Count", fill = "Struck State", title = "Fam Accused") +
 scale_fill_discrete(labels = c("No", "Yes"))

Death <- ggplot(apm, aes(x = is_black, fill = struck_state)) +
geom_bar(position = "fill") +
facet_wrap(~ death_hesitation) +
 labs(x = "Is Black", y = "Count", fill = "Struck State", title = "Death") +
 scale_fill_discrete(labels = c("No", "Yes"))

Knowdef <- ggplot(apm, aes(x = is_black, fill = struck_state)) +

```

```

geom_bar(position = "fill") +
facet_wrap(~ know_def) +
 labs(x = "Is Black", y = "Count", fill = "Struck State", title = "KnowDef") +
 scale_fill_discrete(labels = c("No", "Yes"))

SameRace <- ggplot(apm, aes(x = is_black, fill = struck_state)) +
geom_bar(position = "fill") +
facet_wrap(~ same_race) +
 labs(x = "Is Black", y = "Count", fill = "Struck State", title = "SameRace") +
 scale_fill_discrete(labels = c("No", "Yes"))

FamLaw <- ggplot(apm, aes(x = is_black, fill = struck_state)) +
geom_bar(position = "fill") +
facet_wrap(~ fam_law_enforcement) +
 labs(x = "Is Black", y = "Count", fill = "Struck State", title = "FamLaw") +
 scale_fill_discrete(labels = c("No", "Yes"))

(Accused + Fam_Accused + Death + Knowdef + SameRace + FamLaw) + plot_layout(guides =
"collect")
```

```{r, include=FALSE}
official_apm_reg <- glm(data = apm, struck_state_bin ~ same_race + fam_law_enforcement +
death_hesitation + know_def + accused + is_black * fam_accused, family = binomial)

summary(official_apm_reg)
```

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**Results: **

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We do not believe that the model presented by the APM is adequate. In examining the graph of the proportion of people struck dependent on whether they were black and if they have a friend or family member accused of engaging in criminal activity, we found there to be evidence of an interaction between being black and having a friend or family member who has been accused of criminal activity. This interaction was verified through individual t-tests of the coefficients which provided the p values 0.028081 and 0.021352 respectively—providing strong evidence of a non zero interaction coefficient for both at a 5% significance level. Below is a table showing the estimated coefficients and p values for the predictor variables in our model. The p values provide evidence of non-zero coefficients, the estimates provide evidence of each predictors contribution to the odds of being struck, and the standard errors provide information about the standard error of

each coefficient. Both the coefficient estimates and standard error estimates have been unlogged.

```

```{=latex}
\begin{table}[h]
\begin{tabular}{llll}
Coefficients: & & & \\
& & & \\
Estimate Std. Error z value Pr(>|z|) & Exponentiated Estimate & Exponentiated \\
Std. Error & p value & & \\
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& \textless 2e-16 & & \\
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& 2.28e-14 & & \\
Black + Has an Accused Family/Friend & & 2.16820579 & 1.44947294 \\
& 0.037092 & & \\
\end{tabular}
\end{table}
```

```

For our model we chose to include the same race variable instead of the defendant race variable as it looks like including the defendant race variable does not improve our model since the individual p-values calculated from the t-tests for the 3 defendant race variables are 0.176801, 0.842523 and 0.367357. These insignificant p-values give us the evidence that these variables do not improve our model and therefore, we maintained the same race variable in our model.

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```

```{r, echo = FALSE, fig.height = 4, fig.align='center'}
Fam_Accused
```

```

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```
```{r, include = FALSE}
exp(1.8089)
exp(1.8089 + qnorm(0.975) * (0.1465))
exp(1.8089 - qnorm(0.975) * (0.1465))
vcov(official_apm_reg)
sqrt(0.0214529369 + 0.137807106 + 2*(-0.0139025607))

exp((0.7739 + 1.8089) - qnorm(0.975) * (0.3625671))
exp((0.7739 + 1.8089) + qnorm(0.975) * (0.3625671))
exp(0.7739 + 1.8089)
```
```

****R Appendix: ****